

L8b: Utility-Based Allocation and Bets on the Future

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Objectives for Today

By the end of this lecture, you should be able to:

- Solve budget-constrained utility problems: solve the Cobb–Douglas share-count problem in closed form with SIM preference weights, and state how the CES elasticity moves the allocation.
- Choose a point on the capital allocation line: derive the utility-maximizing risk-free fraction and infer a client's risk aversion.
- Describe a bet on a future condition: write it as an abstract asset, compute its expected NPV, and explain why a risk-averse buyer pays less.

Lectures and Examples

Lecture: Utility-Based Allocation and Bets on the Future

Example 1: Build a Maximum-Utility Portfolio Allocator
Which firms, how much of each, and how does re-solving through 2025 compare with buy-and-hold?

Interview: Measure Risk Aversion with an Income-Gamble Interview
Would you take a job that could double your income or cut it by a third?

Example 2: Choose the Optimal Point on the Capital Allocation Line
Which risk-free fraction maximizes utility, and what A does each answer imply?

Both examples use the default firms unless you uncomment the ticker-file lines.

Concept Review: Utility Functions

Utility functions rank a decision-maker's choices, outcomes, or states of the world. A larger utility is always better.

Utility function: an agent chooses from alternatives $\mathcal{X} = \{x_1, \ldots, x_n\}$. $U : \mathcal{X} \to \mathbb{R}$ assigns each one a number, in *utils*, that ranks the agent's preferences:

$$U(x_i) > U(x_j) \iff x_i \succ x_j \qquad \text{and} \qquad U(x_i) = U(x_j) \iff x_i \sim x_j$$

- Ordinal: U ranks alternatives but does not measure how much better one is, so any order-preserving transformation, e.g., $\ln U$ for positive U, describes the same preferences.
- Key idea: a rational decision-maker always chooses the alternative with the largest utility.

Common Utility Functions

Utility functions, and their parameters, are subjective. Five common forms for a bundle of goods $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \geq 0$ and $\alpha_i > 0$:

Utility function	Form	Models
Linear	$U(\mathbf{x}) = \sum_i \alpha_i x_i$	goods that are perfect substitutes
Logarithmic ($\beta > 0$)	$U(\mathbf{x}) = \ln(\sum_i \alpha_i x_i + \beta)$	diminishing satisfaction from more goods
Cobb–Douglas	$U(\mathbf{x}) = \prod_i x_i^{\alpha_i}$	goods held together
Leontief	$U(\mathbf{x}) = \min(x_1/\alpha_1, \dots, x_n/\alpha_n)$	perfect complements, fixed proportions
CES ($\eta > 0$, $\eta \neq 1$)	$U(\mathbf{x}) = \left(\sum_i \alpha_i x_i^{(\eta-1)/\eta}\right)^{\eta/(\eta-1)}$	substitution set by the elasticity η

We use the Cobb–Douglas and CES utilities next. As η changes, CES moves between the linear ($\eta \to \infty$), Cobb–Douglas ($\eta \to 1$, with $\sum_i \alpha_i = 1$), and Leontief ($\eta \to 0$) forms.

Utility over Holdings

In L7a, every investor held the same tangent portfolio and chose only a risk-free fraction. Instead, ask: given a budget and today's prices, which holdings does the investor prefer most?

The investor chooses share counts that maximize a holding utility. It has no variance term. Risk enters through preference weights that depend on the SIM betas and the state of the market.

Cobb–Douglas Utility

The exponents are the preference weights $\gamma_i \in (-1, 1)$. Maximize the product over every asset, subject to the budget and a floor $n_{\min} > 0$ on each holding:

$$\begin{aligned} &\underset{n_1,\dots,n_{|\mathcal{P}|}}{\text{maximize}} && U(n_1, \dots, n_{|\mathcal{P}|}) = \prod_{i \in \mathcal{P}} n_i^{\gamma_i} \\ &\text{subject to} && \sum_{i \in \mathcal{P}} n_i \, S_i(t) + \text{cash}(t) = W_{\mathcal{P}}(t) && \text{(budget constraint)} \\ & && n_i \geq n_{\min} \quad \forall \, i \in \mathcal{P}, \qquad \text{cash}(t) \geq 0 && \text{(share floor, no borrowing)} \end{aligned}$$

where n_i is the share count of asset i, $S_i(t)$ its price (USD per share), $W_{\mathcal{P}}(t)$ the wealth, and $\text{cash}(t)$ the uninvested wealth (USD), which is not in U. The floor keeps U finite when $\gamma_i < 0$. The preferred set $\mathcal{A}^+ = \{i : \gamma_i > 0\}$ is the basket, and the rest form $\mathcal{A}^- = \{i : \gamma_i \leq 0\}$.

Cobb–Douglas Utility: Floors and Net Budget

Only the ranking matters, so $\ln U$ has the same maximizer as U. With a non-empty basket, we set the floors and the cash first:

$$\frac{\partial \ln U}{\partial n_k} = \frac{\gamma_k}{n_k} \le 0 \qquad k \in \mathcal{A}^-, \text{ so more shares never raise } \ln U$$

$$n_k^\star = n_{\min} \qquad \text{spend every dollar above the floor on } \mathcal{A}^+$$

$$\text{cash}^\star(t) = 0 \qquad \text{a preferred share raises } \ln U, \text{ and cash does not}$$

Substituting both into the budget constraint leaves the net budget for the preferred set:

$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) + n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) = W_{\mathcal{P}}(t)$$

$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) \equiv W_{\text{adj}} > 0$$

A Lagrange multiplier λ then splits W_{adj} over the preferred set, assuming each preferred share count stays above its floor.

Cobb–Douglas Utility: Allocation

Adding the multiplier λ for the net budget gives:

$$\mathcal{L} = \textstyle\sum_{i\in\mathcal{A}^+} \gamma_i \ln n_i - \lambda\left(\sum_{i\in\mathcal{A}^+} n_i S_i(t) - W_{\text{adj}}\right)$$ floor terms are constant

$$0 = \gamma_i/n_i - \lambda\, S_i(t)$$ differentiate with respect to n_i

$$n_i S_i(t) = \gamma_i/\lambda$$ dollars in proportion to γ_i

$$W_{\text{adj}} = \textstyle\sum_{j\in\mathcal{A}^+} \gamma_j/\lambda$$ sum over the preferred set

$$1/\lambda = W_{\text{adj}} \Big/ \textstyle\sum_{j\in\mathcal{A}^+} \gamma_j$$ solve for the multiplier

Substituting $1/\lambda$ into $n_i S_i(t) = \gamma_i/\lambda$ gives:

$$n_i^\star = \underbrace{\frac{\gamma_i}{\sum_{j\in\mathcal{A}^+}\gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \qquad (i\in\mathcal{A}^+), \quad n_i^\star = n_{\min} \quad (i\in\mathcal{A}^-)$$

If a preferred count falls below $n_{\min}$, pin it at the floor, subtract its cost from W_{adj}, and solve again. If every preferred count stays above its floor, dollars are proportional to γ_i.

CES Utility: A Concentration Dial

CES adds the elasticity $\eta > 0$, which sets how willingly the allocator trades one preferred asset for another. For $\eta \neq 1$:

$$U_{\mathrm{CES}} = \Big(\sum_{i \in \mathcal{A}^+} \gamma_i \, n_i^{(\eta-1)/\eta} \Big)^{\eta/(\eta-1)}$$

$$n_i^\star = W_{\mathrm{adj}} \, \frac{\big(\gamma_i / S_i(t)\big)^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t) \big(\gamma_j / S_j(t)\big)^\eta} \qquad (i \in \mathcal{A}^+)$$

Share counts above the floors are proportional to the *bang for the buck* $\gamma_i / S_i(t)$ raised to η. Three limits:

- Unit elasticity ($\eta \to 1$): the Cobb–Douglas allocation.
- Large elasticity ($\eta \to \infty$): the unique largest $\gamma_i / S_i(t)$ gets the whole budget above the floors.
- Small elasticity ($\eta \to 0^+$): equal share counts of every preferred asset.

Proofs: CES companion, including why CES dollar weights depend on share prices when $\eta \neq 1$.

Preference Weights from the Single Index Model

Each trading day, from the SIM parameters (α_i, β_i), the recent market growth rate $\tilde{g}_{M,t}$, and the market-state signal ξ_t:

$$\gamma_i(t) = \tanh\left(h \left[\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t}\, \tilde{g}_{M,t} \right] \right)$$

Here $\tilde{g}_{i,t} = \alpha_i + \beta_i \tilde{g}_{M,t}$ is the SIM's expected growth today, $h = 1$ year makes the argument dimensionless, and the model assumes $\beta_i > 0$.

- Sign picks the basket: $\gamma_i(t) > 0$ when $\tilde{g}_{M,t} > -\alpha_i/\beta_i$. As $\tilde{g}_{M,t}$ rises, the basket grows. As it falls, only the lowest-threshold assets stay. An empty basket holds the floors plus cash.
- Magnitude splits the budget: bearish ($\xi_t > 0$) tilts it toward low-beta assets, bullish toward high-beta.

Exponential Moving Averages

Both market inputs come from the SPY price through exponential moving averages (EMAs). An EMA is a running average that gives recent prices the most weight. For a price series S_t and a window of L trading days:

$$\bar{S}_t = \omega\, S_t + (1-\omega)\, \bar{S}_{t-1} \qquad \text{where} \qquad \omega = \frac{2}{L+1}$$

The Market-State Signal

With $L_{\text{short}} = 21$ and $L_{\text{long}} = 63$ days and a gain $G > 0$, the signal is the scaled crossover of the two price EMAs:

$$\xi_t = -G \left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1 \right)$$

The ratio says where recent prices sit against the quarter's average:

- $\bar{S}_t^{\text{short}} > \bar{S}_t^{\text{long}}$: $\xi_t < 0$, bullish. The market has been rising.
- $\bar{S}_t^{\text{short}} = \bar{S}_t^{\text{long}}$: $\xi_t = 0$, neutral, no tilt. A flat market or an EMA crossing.
- $\bar{S}_t^{\text{short}} < \bar{S}_t^{\text{long}}$: $\xi_t > 0$, bearish. The market has been falling.

The Recent Market Growth

We choose G so that $|\xi_t| \leq 1$ on 99% of training days. The second input is the recent market growth rate:

- $\tilde{g}_{M,t}$ is the 252-day EMA of the daily market growth rates $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$ (inverse years).
- Both inputs on day t use prices through day $t-1$, so no decision uses a price it could not have known.

Example 1: Build a Maximum-Utility Portfolio Allocator

Mean-Variance Utility on the CAL

A complete portfolio c holds w_f in the risk-free asset and $1 - w_f$ in the tangent portfolio $\mathcal{T}$:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_\mathcal{T}] - g_f) \qquad \text{expectation is linear}$$

$$\sigma^2_{g,c} = (1 - w_f)^2 \, \sigma^2_{g,\mathcal{T}} \qquad \text{the risk-free asset adds no variance}$$

Lending ($0 < w_f < 1$) and borrowing ($w_f < 0$) move c along the capital allocation line, whose slope is the Sharpe ratio of $\mathcal{T}$.

By two-fund separation, investors with the same inputs who can lend and borrow freely at g_f hold the same $\mathcal{T}$, so preferences set only w_f.

Which w_f should a given investor choose?

Certainty Equivalent and Risk Premium

Recall from L7b: we rank uncertain choices by expected utility $\mathbb{E}[U(W)]$, and a concave U means risk aversion:

$$\underbrace{U(\mathrm{CE})}_{\text{utility of the sure wealth}} = \underbrace{\mathbb{E}[U(W)]}_{\text{expected utility of the risky outcome}} \qquad \mathrm{RP} = \underbrace{\mathbb{E}(W)}_{\text{expected wealth}} - \underbrace{\mathrm{CE}}_{\text{sure wealth}}$$

$$\mathrm{CE} = U^{-1}(\mathbb{E}[U(W)])$$

For example, let $U(W) = \ln W$, and flip a fair coin for a wealth of 50 or 150 USD:

$\mathbb{E}(W) = \frac{1}{2}(50) + \frac{1}{2}(150) = 100$	average wealth, USD
$U(\mathbb{E}(W)) = \ln 100 \approx 4.6052$	utility of the sure average, utils
$\mathbb{E}[U(W)] = \frac{1}{2}\ln 50 + \frac{1}{2}\ln 150 \approx 4.4613$	expected utility of the gamble, utils
$\mathrm{CE} = U^{-1}(\mathbb{E}[U(W)]) = e^{4.4613} \approx 86.60$	$U^{-1}(x) = e^x$ undoes $\ln$, USD
$\mathrm{RP} = 100 - 86.60 = 13.40$	risk premium, USD

The 50 USD gained adds less utility than the 50 USD lost takes away, so the investor takes 86.60 USD for certain over a flip worth 100 USD on average.

Arrow–Pratt Measures

How risk averse is a decision-maker? Arrow and Pratt (1964) measure the curvature of a utility of wealth with $U'(w) > 0$:

$$\underbrace{r(w) = -\frac{U''(w)}{U'(w)}}_{\text{absolute}} \qquad \underbrace{\bar{r}(w) = w\, r(w)}_{\text{relative}}$$

	$U(w)$	$r(w)$	$\bar{r}(w)$
Log	$\ln w$	$1/w$	1
Linear	$a\,w + b$	0	0
Power	w^τ/τ	$(1-\tau)/w$	$1-\tau$

$r > 0$: averse. $r = 0$: neutral. $r < 0$: seeking.

For a small risk, larger r means a lower CE:

$$\mathrm{RP} \approx \tfrac{1}{2}\, r(\mathbb{E}(W))\ \mathrm{Var}(W)$$

For a portfolio: with small returns, CE return ≈ expected return $-\ \frac{\bar{r}}{2} \times$ return variance. Mean-variance utility has this form, with A in the role of $\bar{r}$.

Mean-Variance Utility

We choose w_f by maximizing the investor's utility. Mean-variance utility rises with expected growth and falls with variance:

$$\boxed{U(w_f) = \mathbb{E}[g_c] - \frac{A}{2}\,\Delta t\,\sigma^2_{g,c}}$$

The risk-aversion coefficient $A > 0$ (dimensionless) sets how much the investor dislikes variance. As in L5a, $\Delta t\,\sigma^2_{g,c}$ is the squared volatility, in year^{-1}. On the line, both terms depend only on w_f:

$$U(w_f) = g_f + (1 - w_f)(\mathbb{E}[g_{\mathcal{T}}] - g_f) - \tfrac{A}{2}(1 - w_f)^2\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{substitute the line}$$

$$\frac{dU}{dw_f} = -\big(\mathbb{E}[g_{\mathcal{T}}] - g_f\big) + A\,(1 - w_f)\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{differentiate}$$

$$0 = -\big(\mathbb{E}[g_{\mathcal{T}}] - g_f\big) + A\,(1 - w^\star_f)\,\Delta t\,\sigma^2_{g,\mathcal{T}} \qquad \text{first-order condition}$$

The Optimal Complete Portfolio

Solving the first-order condition for the fraction in $\mathcal{T}$ gives:

$$1 - w_f^\star = \frac{\mathbb{E}[g_\mathcal{T}] - g_f}{A\,\Delta t\,\sigma_{g,\mathcal{T}}^2}$$

- A maximum: the second derivative, $-A\,\Delta t\,\sigma_{g,\mathcal{T}}^2$, is negative.
- What moves it: more excess growth raises the fraction in $\mathcal{T}$. A larger A or $\sigma_{g,\mathcal{T}}^2$ lowers it, and doubling A halves it.
- Who lends and who borrows? The investor with $A_\mathcal{T} = (\mathbb{E}[g_\mathcal{T}] - g_f)/(\Delta t\,\sigma_{g,\mathcal{T}}^2)$ holds only $\mathcal{T}$. $A > A_\mathcal{T}$ lends, and $A < A_\mathcal{T}$ borrows to buy more of $\mathcal{T}$.

Indifference Curves and Risk Aversion

An indifference curve joins the portfolios with the same utility. Fixing U gives:

$$\mathbb{E}[g_c] = U + \frac{A}{2}\,\Delta t\,\sigma_{g,c}^2$$

Against $\sigma_{g,c}$, each curve is a parabola that opens upward, and higher curves are better. The optimum is where the highest reachable curve touches the line.

The result also runs backward. A client who picks $w_f < 1$ implies:

$$\boxed{A = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{(1 - w_f)\,\Delta t\,\sigma_{g,\mathcal{T}}^2}}$$

Measuring A with Income Gambles

Can we measure A without asking for w_f? Barsky et al. (1997) offered two jobs: a sure income, or equal chances of doubling it or cutting it by δ. At the cut $\delta^\star$ where a person cannot choose, in units of the sure income:

$$\boxed{\tfrac{1}{2}\,U(2) + \tfrac{1}{2}\,U(1-\delta^\star) = U(1)}$$

$$\tfrac{1}{2}\ln 2 + \tfrac{1}{2}\ln\tfrac{1}{2} = 0 = U(1) \qquad U(w) = \ln w \;(\bar{r}=1),\; \delta^\star = \tfrac{1}{2}$$

$$\tfrac{1}{2}(-\tfrac{1}{2}) + \tfrac{1}{2}(-\tfrac{3}{2}) = -1 = U(1) \qquad U(w) = -1/w \;(\bar{r}=2),\; \delta^\star = \tfrac{1}{3}$$

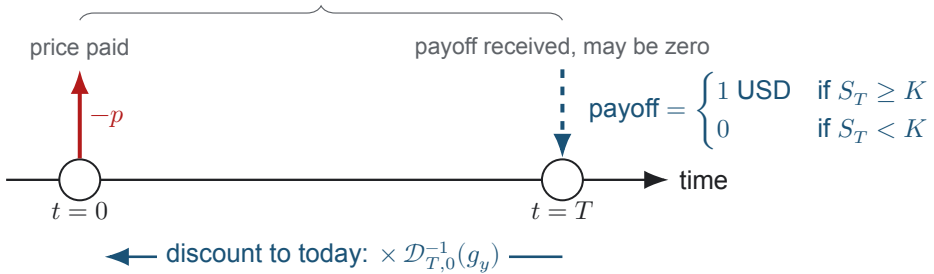
More risk-averse people accept only smaller cuts, so a few questions bracket $\bar{r}$, and with it A.

Interview and Example 2: Income-Gamble Interview Choose the Optimal Point on the Capital Allocation Line

Derivatives: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S_T \geq K$ (the L5a trade rule), and nothing otherwise.

a bet: pay today, paid T years from now only if the condition is true



A Bet as an Abstract Asset

In L1b's language, the contract is an abstract asset with two cash flows:

$$V(S_T) = \begin{cases} 1 & \text{if } S_T \geq K \\ 0 & \text{if } S_T < K \end{cases} \qquad \mathrm{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S_T), \quad \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}$$

Unlike a Treasury bill (L2a), this NPV is random: $-p + e^{-g_y T}$ if true, $-p$ if false. A one-or-zero payoff has an expected value equal to the probability that it is one, so under our beliefs:

$$\boxed{\mathbb{E}[\mathrm{NPV}_0] = -p + e^{-g_y T}\, \mathbb{P}(S_T \geq K)}$$

For GBM, $\mathbb{P}(S_T \geq K) = 1 - \Phi(z_\star)$ from L5a. The expected NPV is positive when $p < e^{-g_y T}\, \mathbb{P}(S_T \geq K)$.

What Would a Risk-Averse Buyer Pay?

On its own, the payoff is a gamble, like the coin flip earlier. A risk-averse buyer values it at its certainty equivalent, below its expected value, so pays less than the discounted expected payoff. How much less depends on the buyer's risk aversion and the number of contracts.

Contracts whose payoffs depend on another asset's future price are derivatives.

Each has one market price, yet buyers differ in beliefs and risk aversion.

What sets the price?

Optional Advanced Material

The advanced index describes the optional notebook:

- CES limits and log-linear utility: derive the closed-form CES optimum and prove its three limits. Show that log-linear utility has the Cobb–Douglas optimizer, and why CES weights depend on share price levels.

Summary

- The single index model and Cobb–Douglas utility turn preferences into holdings: signs pick the basket, and Cobb–Douglas splits the preferred budget by the positive weights. The CES elasticity moves the split toward concentration or equal share counts.
- Risk aversion picks the point on the capital allocation line: measured with the certainty equivalent and Arrow–Pratt, it sets the risky fraction in closed form. A client's chosen fraction implies the coefficient that income gambles bracket.
- A bet on the future is an abstract asset with a random cash flow: its expected NPV depends on our beliefs, and a risk-averse buyer of the bet alone pays less.

Preference weights pick the assets, risk aversion picks the point on the capital allocation line, and a bet on the future is an abstract asset.